Run report: pkm_supersingular_aux, 2026-10-15T23:15Z

research/pkm_supersingular_aux_20261015 · observations only; interpretation is in `../../README.md`

What ran. `run_q3.sh` (ecbench sessions s20, s22, s24), `run_q2.sh` (F4 cells) and `run_q1.sh` (structure), in that order, from commit `8d719fddf` (`PROTOCOL.md`), binaries in `binaries.sha256` (`ecbench 92e39467…`, `pkm_ss_tower f277e003…`), on a shared 4-vCPU, 15 GiB container: one thread, `nice -n 10`, 4 GB address-space cap, ecbench at L0 (`--cpus none --allow-busy`). Start 2026-10-15T23:15:13Z; every start and end is in `progress.txt`.

Failures. A container restart killed `run_q2.sh` after 23:26:36Z, during `T31-ordinary-m34` (8 partial rows kept in `q2/interrupted/`). The script was made resumable (completed runs skipped) and restarted; that run was repeated whole. `run_q1.sh` ran in resumable chunks of 300 in `p`. No system timed out or stopped for size; no recount mismatched.

Q3 (measured). 3 sessions × 168 records (56 warm-up): 504/504 verified; `verify --replay-all` reproduced 112 replays per session, audit OK ×3. Mean S and paired ratios are in `q3/tables/summary-arms.tsv` and `summary-ratios.tsv` (ecbench's ratios and 95 % cluster-bootstrap intervals, 16 pairs on 8 workloads).

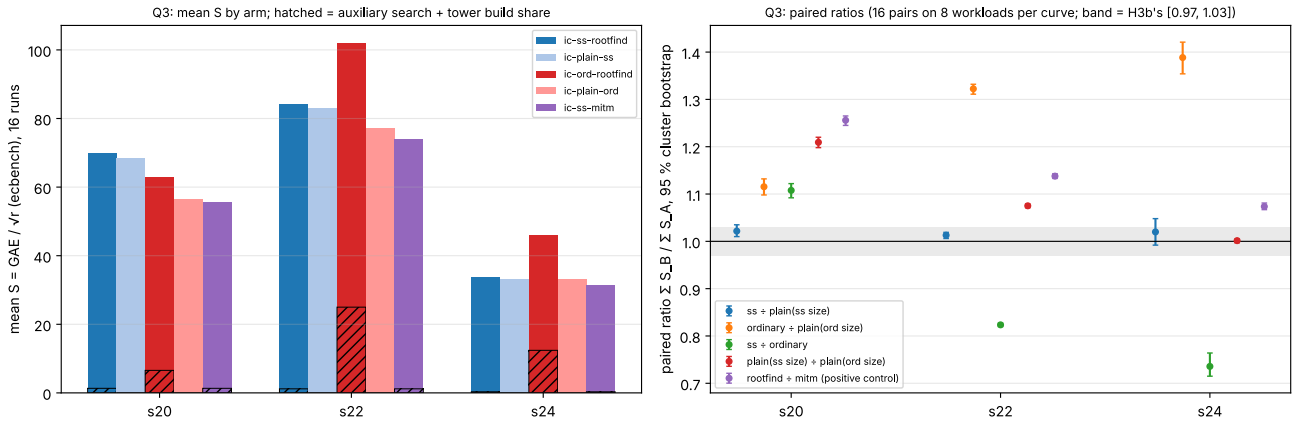


Figure 1: Q3: mean S per arm and paired ratios, sessions `q3/sessions/s20`, `s22`, `s24`.

Q2 (measured). 336 systems in 24 run files (`q2/runs/*.jsonl`, `q2/cells.tsv`), two random targets per cell, all inconsistent (no decomposition), 296 with an exhaustive recount that agreed.

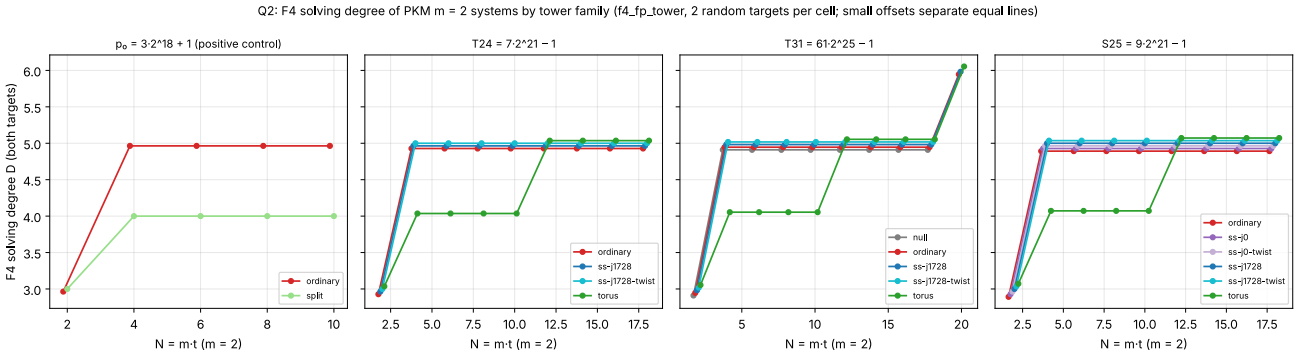


Figure 2: Q2: F4 solving degree against N at $m = 2$, per prime and tower family.

Q1 (exact). 10 chunks, 103 primes $p \equiv 3 \bmod 4$ in $[103, 2999]$, 195 (p, t) cells, 12,581 rows (`q1/structure.jsonl`); brute-force agreement 906/906 at $p \leq 131$; positive control 100 % in every cell. Chunk times 14 s to 23 min (the first carries the brute force).

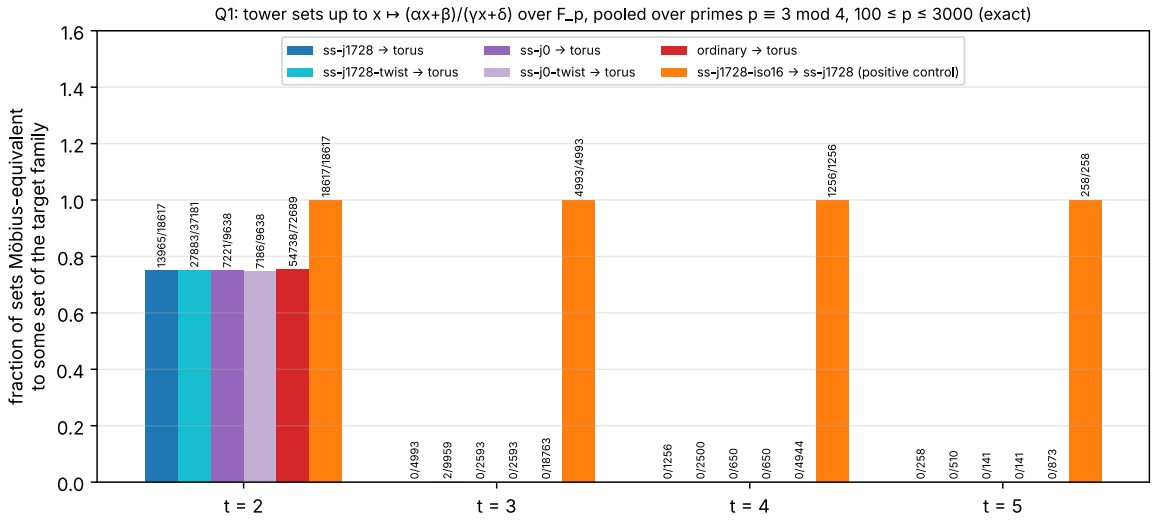


Figure 3: Q1: fraction of sets Möbius-equivalent to a torus set (or, for the positive control, to a $y^2 = x^3 + x$ set), pooled over primes.